Volumes by Known Cross Sections and Shells

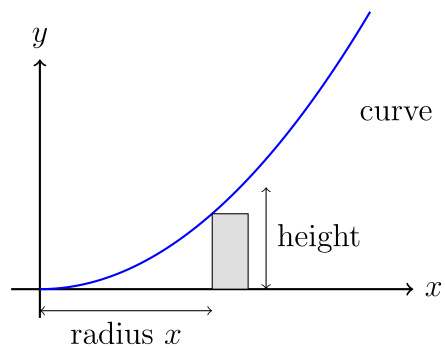
AP Calculus AB/BC Applications of Integration

Volumes by integrating a known cross-sectional area, volumes by cylindrical shells, and choosing between the two

The two set-ups.

- **Known cross sections:** $V = \int_a^b A(x)\,dx$, where $A(x)$ is the area of the slice at x. Square: $A = s^2$. Semicircle on a diameter d: $A = \frac{\pi}{8}d^2$. Equilateral triangle of side s: $A = \frac{\sqrt{3}}{4}s^2$.
- **Cylindrical shells** (revolving about the y-axis): $V = \int_a^b 2\pi\,(\text{radius})(\text{height})\,dx$, with radius x and height the vertical extent of the region at x.
- Give exact answers unless a decimal is asked for, and state the units of the volume.

The shell picture used throughout. A vertical strip of the region, at distance x from the axis, sweeps out a thin cylinder when the region turns. No values are shown here — use the functions given in each problem.



1. The base of a solid is the region bounded by $y = x$, the x-axis, and the line $x = 4$. Every cross section perpendicular to the x-axis is a square whose side lies in the base.
Set up and evaluate an integral for the volume of the solid.

Answer: ________________ cubic units

2. The base of a solid is the region bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 9$. Every cross section perpendicular to the x-axis is a semicircle whose *diameter* lies in the base.
Find the exact volume of the solid.

Answer: ________________ cubic units

3. The region under $y = x^2$ from $x = 0$ to $x = 3$, above the x-axis, is revolved about the y-axis.
Use the method of cylindrical shells to find the exact volume of the solid of revolution.

Answer: ________________ cubic units

4. A monument sits on a base shaped like the region bounded by $y = \sqrt{16 - x^2}$ and the x-axis. Every cross section perpendicular to the x-axis is a square whose side lies in the base.
Find the exact volume of the monument.

Answer: ________________ cubic units

5. The region under $y = \sqrt{x}$ from $x = 0$ to $x = 4$, above the x-axis, is revolved about the y-axis.
Use cylindrical shells to find the exact volume. (Solving for x in terms of y is exactly the work shells let you skip.)

Answer: ________________ cubic units

6. The base of a solid is the region bounded by $y = 4 - x^2$ and the x-axis. Every cross section perpendicular to the x-axis is an equilateral triangle with one side in the base.
Find the exact volume, then give it to two decimal places.

Answer: ________________ cubic units

7. The region bounded by $y = 2x$ and $y = x^2$ is revolved about the y-axis.
First find the x-values where the two curves meet, then use cylindrical shells to find the exact volume. Be careful about which curve gives the height of a shell.

Answer: ________________ cubic units

8. The region bounded by $y = x^2$, the y-axis, and the line $y = 4$ is revolved about the y-axis.
Explain in one or two sentences why cylindrical shells in x is a reasonable choice for this region, then set up the shell integral and evaluate it exactly.

Answer: ________________ cubic units